

SKYNET ALGORITHM FOR SINGLE-DISH RADIO MAPPING I: CONTAMINANT-CLEANING, MAPPING, AND PHOTOMETERING SMALL-SCALE STRUCTURES

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ABSTRACT

We present a single-dish mapping algorithm with a number of advantages over traditional techniques. (1) Our algorithm makes use of weighted modeling, instead of weighted averaging, to interpolate between signal measurements. This smooths the data, but without blurring the data beyond instrumental resolution. Techniques that rely on weighted averaging blur point sources sometimes as much as 40%. (2) Our algorithm makes use of local, instead of global, modeling to separate astronomical signal from instrumental and/or environmental signal drift along the telescope's scans. Other techniques, such as basket weaving, model this drift with simple functional forms (linear, quadratic, etc.) across the entirety of scans, limiting their ability to remove such contaminants. (3) Our algorithm makes use of a similar, local modeling technique to separate astronomical signal from radio-frequency interference (RFI), even if only continuum data are available. (4) Unlike other techniques, our algorithm does not require data to be collected on a rectangular grid or regridded before processing. (5) Data from any number of observations, overlapping or not, may be appended and processed together. (6) Any pixel density may be selected for the final image. We present our algorithm, and evaluate it using both simulated and real data. We are integrating it into the image-processing library of the Skynet Robotic Telescope Network, which includes optical telescopes spanning four continents, and now also Green Bank Observatory's 20-meter diameter radio telescope in West Virginia. Skynet serves hundreds of professional users, and additionally tens of thousands of students, of all ages. Default data products are generated on the fly, but will soon be customizable after the fact.

Subject headings: techniques: image processing — radio continuum: general

1. INTRODUCTION

1.1. *Skynet*

Founded in 2005, Skynet is a global network of fully automated, or robotic, volunteer telescopes, scheduled through a common web interface.⁷ Currently, our optical telescopes range in size from 14 to 40 inches, and span four continents. Originally envisioned for gamma-ray burst follow-up (Reichart et al. 2005, Haislip et al. 2006, Dai et al. 2007, Updike et al. 2008, Nysewander et al. 2009, Cenko et al. 2011, Cano et al. 2011, Bufano et al. 2012, Jin et al. 2013, Morgan et al. 2014, Martin-Carrillo et al. 2014, Friis et al. 2015, De Pasquale et al. 2016, Bardho et al. 2016, Melandri et al. 2017), Skynet has also been used to study gravitational-wave sources (Abbott et al. 2017a, 2017b, Valenti et al. 2017, Yang et al. 2017), blazars (Osterman Meyer et al. 2008, Valtonen et al. 2016, Zola et al. 2016, Liu et al. 2017, Goyal et al. 2017), supernovae (Foley et al. 2010, 2012, 2013,

Pignata et al. 2011, Valenti et al. 2011, 2014, Pastorello et al. 2013, Milisavljevic et al. 2013, Maund et al. 2013, Fraser et al. 2013, Stritzinger et al. 2014, Inserra et al. 2014, Takats et al. 2014, 2015, 2016, Dall'Ora et al. 2014, Folatelli et al. 2014, Barbarino et al. 2015, de Jaeger et al. 2016, Gutierrez et al. 2016, 2018, Tartaglia et al. 2017, 2018, Prentice et al. 2018), supernova remnants (Trotter et al. 2017), novae (Schaefer et al. 2011), pulsating white dwarfs and hot subdwarfs (Thompson et al. 2010, Barlow et al. 2010, 2011, 2013, 2017, Reed et al. 2012, Bourdreaux et al. 2017, Hutchens et al. 2017), a wide variety of variable stars (Layden et al. 2010, Gvaramadze et al. 2012, Wehrung et al. 2013, Miroshnichenko et al. 2014, Abbas et al. 2015, Khokhlov et al. 2017, 2018), a wide variety of binary stars (Reed et al. 2010, Sarty et al. 2011, Helminiak et al. 2011, 2012, 2015, Strader et al. 2015, Tovmassian et al. 2016, 2017, Fuchs et al. 2016, Pala et al. 2017, Neustroev et al. 2017, Lubin et al. 2017, Swihart et al. 2017, Zola et al. 2017, Kriwattanawong et al. 2018), exoplanetary systems (Fischer et al. 2006, Czesla et al. 2012, Meng et al. 2014, 2015, Kenworthy et al. 2015, Kuhn et al. 2016, Awiphan et al. 2016, Blank et al. 2018), trans-Neptunian objects and Centaurs (Braga-Ribas et al. 2013, 2014, Dias-Oliveira et al. 2015), asteroids (Descamps et al. 2009, Pravec et al. 2010, 2012, 2016, Marchis et al. 2012, Savanevych et al. 2018), and near-Earth objects (NEOs; Brozovic et al. 2011, Pravec et al. 2014). Skynet is also the leading tracker of NEOs in the southern hemisphere (R. Holmes,

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⁷ <https://skynet.unc.edu>



FIG. 1.— Green Bank Observatory 20-meter diameter radio telescope. (Photo credit: GBO)

private communication).

Skynet’s mission is split evenly between supporting professional astronomers and supporting students and the public. Although most of our observations have been for professionals, most of our users are students. We have developed/are continuing to develop Skynet-based curricula for undergraduates, high-school students (in partnership with Morehead Planetarium and Science Center (MPSC)), and middle school-aged students (in partnership with the University of Chicago/Yerkes Observatory, Green Bank Observatory (GBO), the Astronomical Society of the Pacific, and 4-H), as well as for blind and visually-impaired students (in partnership with Associated Universities, Inc., the University of Chicago/Yerkes Observatory, the Technical Education Research Centers, and the University of Nevada at Las Vegas). These efforts have reached over 20,000 students, and our public-engagement efforts (also in partnership with MPSC) have also reached over 20,000, mostly elementary and middle school-aged students. Curriculum-based student users queue observations through the same web interface that the professionals use. Altogether, over 15 million images have been taken to date.

In partnership with GBO, and funded by the American Recovery and Reinvestment Act, Skynet has added its first radio telescope, GBO’s 20-meter in West Virginia (see Figure 1). We describe the 20-meter, which has been refurbished, in §2. As with Skynet’s optical telescopes, the 20-meter serves both professionals and students. Professional use consists primarily of timing observations (e.g., pulsar timing; fast radio burst searches in conjunction with *Swift*), but also some mapping observations, for photometry (e.g., fading of Cas A and improved flux-density calibration of the radio sky (Trotter et al. 2017); intra-day variable blazar campaigns in conjunction with other radio, and optical, telescopes). Student use consists of timing, spectroscopic (e.g., Williamson et al. 2018), and mapping observations, but with an emphasis on mapping, at least for beginners.

Regarding student, as well as public, use, the 20-meter represents a significant opportunity for radio astronomy. Small optical telescopes can be found on many, if not most, college campuses. But small radio telescopes are significantly more expensive to build, operate, and main-

tain, and consequently are generally found only in the remote locations that make the most sense for professional use. Consequently, most people – including most students of astronomy – never experience radio telescopes, let alone use them. However, under the control of Skynet, the 20-meter is not only more accessible to more professionals, it is already being used by thousands of students per year, of all ages, as well as by the public.

1.2. Single-Dish Mapping

In this paper, we present the single-dish mapping algorithm that we developed for Skynet, both for professional use and for student use. Here we outline the design requirements that we set for ourselves, and outline approaches that we adopted/developed to meet these requirements.

1.2.1. Mapping Pattern

Many single-dish mapping algorithms (e.g., Sofue & Reich 1979, Emerson & Grave 1988) require the signal to be sampled on a rectangular grid. Generally, this requires what is called “step-and-integrate” or “point-and-shoot” mapping.

However, this is an inefficient way to observe, with greater telescope acceleration, deceleration, and settling overheads, greater wear and tear on the telescope, and greater sensitivity to time-variable systematics (see §1.2.3). “On-the-fly” (OTF) mapping, in which signal is integrated while the telescope moves, minimizes these concerns, as long as integrations span no more than ≈ 0.2 beamwidths (FWHM) along the telescope’s direction of motion (along the telescope’s scan), to avoid blurring point sources by more than $\approx 1\%$ (Mangum, Emerson & Greison 2007).

A wide variety of OTF mapping patterns might be employed (e.g., see Figure 2):

- A “raster” pattern approximates a rectangular grid, though integrations might not line up from scan to scan.
- A “nodding” pattern, in which the telescope moves up and down, typically in elevation, as Earth’s rotation carries the telescope’s beam across the sky in the perpendicular direction, is typical of meridian-transit telescopes, which cannot move east-west (e.g., see §2.2). However, even with full-motion telescopes, a nodding pattern can be used to maintain a constant parallactic angle, if it is desired that the telescope’s beam pattern not rotate across the image, or from image to image.
- “Spiral” and “hypocycloid” patterns are efficient ways of mapping sources and extended structures, respectively, without abrupt changes to the telescope’s motion (Mangum, Emerson & Greison 2007).
- A “daisy” pattern can also be used to map sources, with the advantage of crossing the source’s peak, and sampling the background level, an arbitrarily large number of times.

As long as the gaps between scans do not exceed Nyquist sampling, or ≈ 0.4 beamwidths, in theory all information between scans can be recovered (modulo noise,

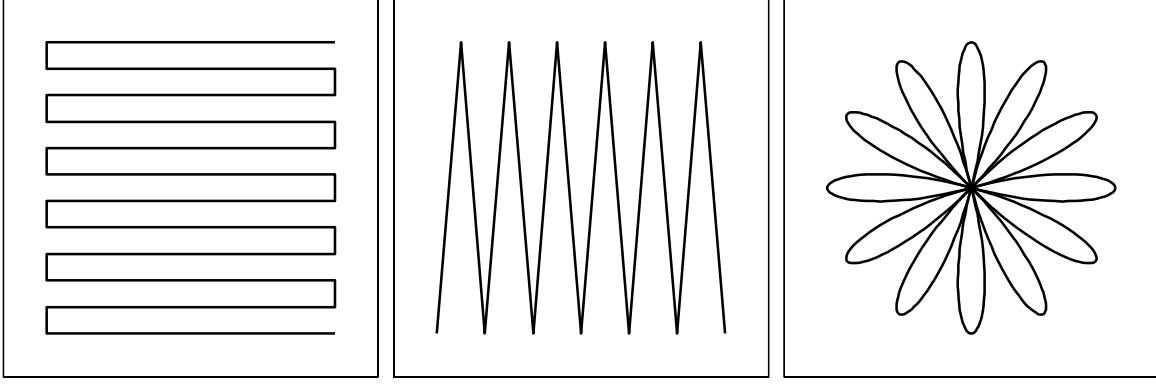


FIG. 2.— OTF mapping patterns. **Left:** Raster (horizontal). **Middle:** Nodding. **Right:** Daisy.

interference, etc.) Given this, and not wishing to limit our users to a particular mapping pattern, or set of patterns, we require that our algorithm work independently of mapping pattern. Furthermore, since student/quick-look maps may be undersampled, we also require that our algorithm work independently of sampling density (within reason). Instead, undersampled images will be flagged as not for professional use.

1.2.2. Signal Averaging vs. Signal Modeling

Some single-dish mapping algorithms get around the problem of not collecting data on a rectangular grid by resampling the data onto a rectangular grid before processing (e.g., Winkel, Floer & Kraus 2012). This is known as “regridding”, and is typically done by taking a weighted average of the data around each grid point. This is essentially a convolution of the data with the weighting function, sometimes called the convolution kernel.

However, this blurs the image. Many adopt a kernel of width equal to the telescope’s beamwidth, but this results in $\approx 40\%$ blurring and $\approx 40\%$ errors (near the center of the beam pattern) in the reconstructed image (see Figure 3). Others oversample so a narrower kernel can be used. For example, Winkel, Floer & Kraus (2012) collect ≈ 3 times as much data as required by Nyquist and use a $1/2$ -beamwidth kernel, but this still results in $\approx 12\%$ blurring and $\approx 18\%$ errors (near the center of the beam pattern) in the reconstructed image (Figure 3).

Given that single-dish maps already suffer from poor resolution (compared to interferometric maps), we do not want to degrade resolution further, simply due to processing. To this end, instead of weighted *averaging*, we use weighted *modeling*: Instead of averaging the data local to each pixel in the final image, using a weighting function, we fit a model to the data local to each pixel in the final image, using a weighting function (see Figure 4). As long as (1) the model is sufficiently flexible over the scale of the weighting function, and (2) sampling is sufficiently dense to constrain (actually, overconstrain) this model, the signal should be recoverable at any location, without blurring. For example, the weighted modeling approach that we present in §3.7 is able to recover the simulated data in Figure 3 with $<1\%$ errors (near the center of the beam pattern).

Another requirement that we place on our algorithm is that this – replacing the data with a locally modeled version of itself – be the last step, not the first step. Other

algorithms regrid the data before processing (e.g., contaminant removal, see §1.2.3), because their processing operations work only on gridded data. However, this is a poor modeling practice: Even if weighted averaging could be done accurately, and even if weighted modeling can be done accurately, every operation on data propagates uncertainty in ways that are both difficult to understand and even more difficult to properly account for before/in the next operation. It is always preferable to operate on real data, at least for as long as possible, than it is to operate on successive approximations thereof.

1.2.3. Contaminant Removal

Single-dish mapping algorithms must address signal contaminants, which we separate into three broad categories: (1) en-route signal drift, resulting in what is sometimes called “the scanning effect”, (2) radio-frequency interference (RFI), and (3) elevation-dependent signal. Each of these is demonstrated in Figure 5.

En-Route Drift: Even with very stable, modern receivers, the detected signal can still drift in time, due to instrumental reasons, such as $1/f$, or pink, noise, and/or due to environmental reasons, such as changing atmospheric emission, or changing ground emission spilling over the edge of the dish, particular as the dish moves. This results in low-level variations in the signal along the telescope’s scans, and is most noticeable from scan to scan (the scanning effect). Components of these variations can be made to vary over shorter or longer angular scales by moving the telescope slower or faster, but this does not eliminate them.

Sofue & Reich (1979) used unsharp masking to separate en-route drift (and small-scale structure in the image) from larger-scale structure, and then modeled the en-route drift along each scan with a second-order polynomial, using sigma-clipping to avoid small-scale structure contamination. Drawbacks of this approach are: (1) Unsharp masking uses a blurred version of the data to correct the data, resulting in a blurred image, at least in the lower signal-to-noise parts of the image (something we wish to avoid; §1.2.2); (2) Low-order polynomials may adequately model en-route drift over small angular scales, but are too simple/increasingly inaccurate over larger angular scales, such as the length of a scan; and (3) Sigma-clipping is too crude of an outlier rejection criterion (see §1.3).

Emerson & Grave (1988) Fourier transform the data,

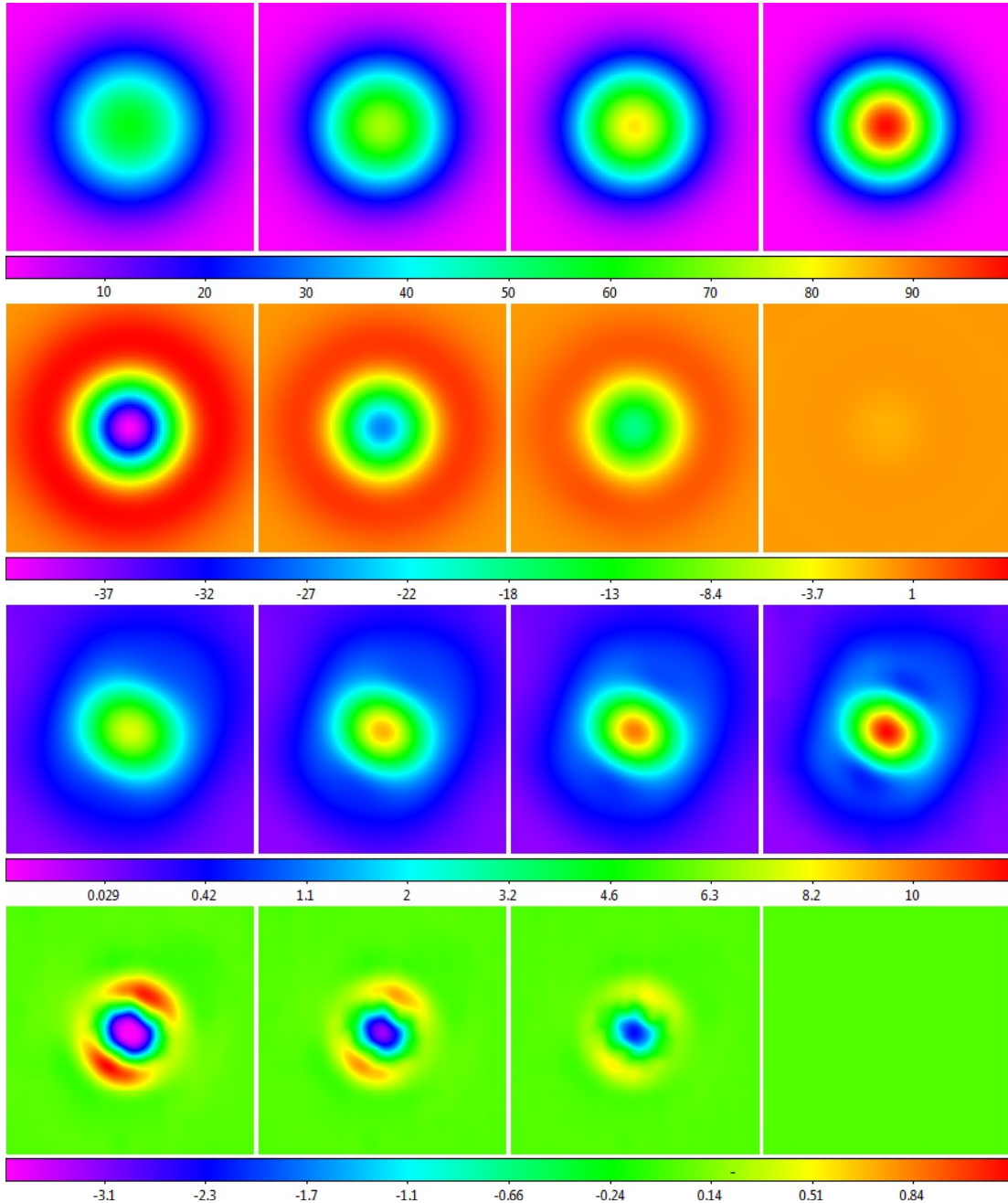


FIG. 3.— **First Row:** Simulation of a point source sampled with a Gaussian beam pattern on a 1/5-beamwidth grid, recovered using, from left to right: 1-beamwidth weighted averaging, 2/3-beamwidth weighted averaging, 1/2-beamwidth weighted averaging, and weighted modeling, as described in §3.7. **Second Row:** Residual error associated with each of these techniques. **Third Row:** Cassiopeia A observed with one of the 20-meter's L-band linear polarization channels using a 1/5-beamwidth raster, and recovered using the above techniques. Weighted averaging fails to recover the telescope's beam pattern, which is structured. **Fourth Row:** Difference between each of these techniques and weighted modeling. Square-root and squared scalings are used in the third and fourth rows, respectively, to emphasize fainter beam structure (units are dimensionless, with one corresponding to the noise diode; see §3.1.)

collapsing en-route drift to a single band of near-zero spatial frequencies, perpendicular to the scan direction. They then mask it in Fourier space and transform back. Ideally, two orthogonal mappings are combined (in Fourier space), so real spatial frequencies that are masked in one image can be recovered from the other. The primary advantage of this approach is that it does not assume that en-route drift can be well-modeled by a low-order polynomial over the length of each scan. However, drawbacks are: (1) We will not always have two orthog-

onal mappings; (2) Even if we did, OTF rasters do not populate rectangular grids, requiring the data to be re-gridded before processing (however, §1.2.2); and (3) This technique would not work well, or at all, with other mapping patterns (e.g., noddings, daisies, etc.; §1.2.1).

Haslam, Quigley & Salter (1970), Haslam et al. (1974), Seiber, Haslam & Salter (1979), and Haslam et al. (1981) introduced a technique called basket-weaving, involving two mappings with intersecting scans (not necessarily orthogonal). Signal differences at the intersections are min-

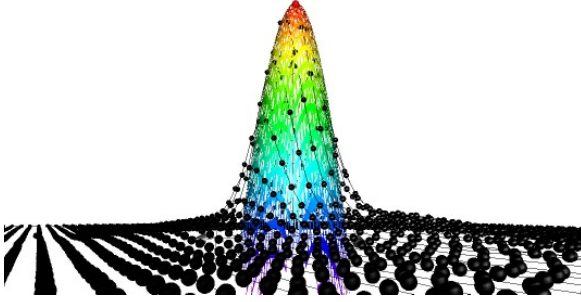


FIG. 4.— Weighted modeling of 20-meter data from the third row of Figure 3, at two representative points. Modeled surfaces span two beamwidths, but are most strongly weighted to fit the data over only the central, typically, $1/3 - 2/3$ beamwidths, as described in §3.7. Only the central point (red) is retained.

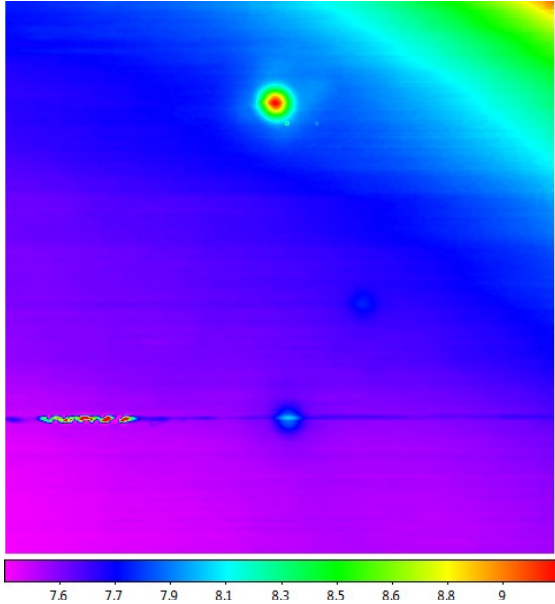
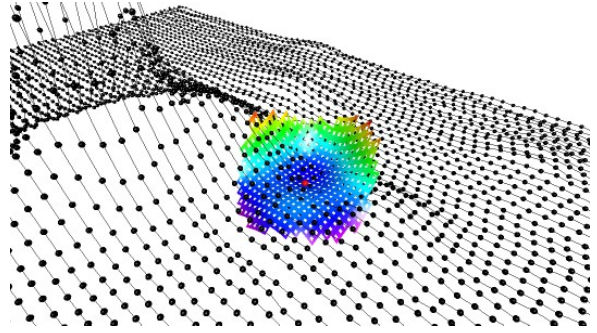


FIG. 5.— Raw map of Virgo A (top), 3C 270 (center right), and 3C 273 (bottom), acquired with the 20-meter in L band, using a $1/10$ -beamwidth horizontal raster. Left and right linear polarization channels have been summed, partially symmetrizing the beam pattern (Figure 3). Locally modeled surface (§1.2.1, see §3.7) has been applied for visualization only. All three signal contaminants are demonstrated: (1) en-route drift, the low-level variations along the horizontal scans, (2) RFI, both long-duration, during the scan that passes through 3C 273, and short-duration, near Virgo A, and (3) elevation-dependent signal, toward the upper right, which was only $\approx 11^\circ$ above the horizon.



FIG. 6.— Green Bank Observatory 40-foot diameter radio telescope. (Photo credit: GBO)

imized, but again assuming that en-route drift can be well-modeled by a low-order polynomial over the length of each scan. The procedure also requires iteration. Winkel, Floer & Krauss (2012) introduced an updated version that does not require iteration, but: (1) It does require regridding (§1.2.2); (2) It works only with two, near-orthogonal mappings, so not with single mappings, and not with noddings, daisies, etc. (§1.2.1); and (3) Although it does permit an arbitrarily high-order parameterization of the en-route drift over the length of each scan, in practice it becomes less effective as the number of parameters times the number of scans approaches the number of regridded pixels (which is typically significantly less than the original number of signal measurements). This limited them to second-order polynomials in the examples that they presented.

RFI: RFI is typically localized to specific frequencies.

If spectral information is available, these frequencies can be identified and masked, or the excess signal at these frequencies can be measured and subtracted off (see §3.1 of Paper II, Dutton et al. 2018). However, sometimes only continuum data are available (e.g., see §2.2), or spectral data are available but the RFI has a continuous spectrum (e.g., lightning). In these cases, RFI can be identified only from its temporal signature. If prolonged, it might be indistinguishable from en-route drift. If localized in time, it is unlikely to occur at the same position on adjoining scans, appearing as a source that is narrower than the telescope’s beamwidth, at least across scans. In most of the above references, RFI was identified by eye and excised by hand. However, given the volume, and diversity, of users that we have with the 20-meter, we need to be able to do this automatically.

Elevation-Dependent Signal: Atmospheric emission increases as elevation decreases. And if the dish is over-illuminated, terrestrial spillover increases as elevation increases. If interested only in small-scale structures, these backgrounds can be removed with en-route drift (see §3.3). However, if large-scale structures are to be retained (or added back in), these backgrounds need to be removed separately (see §2.3 of Paper II, Dutton et al. 2018).

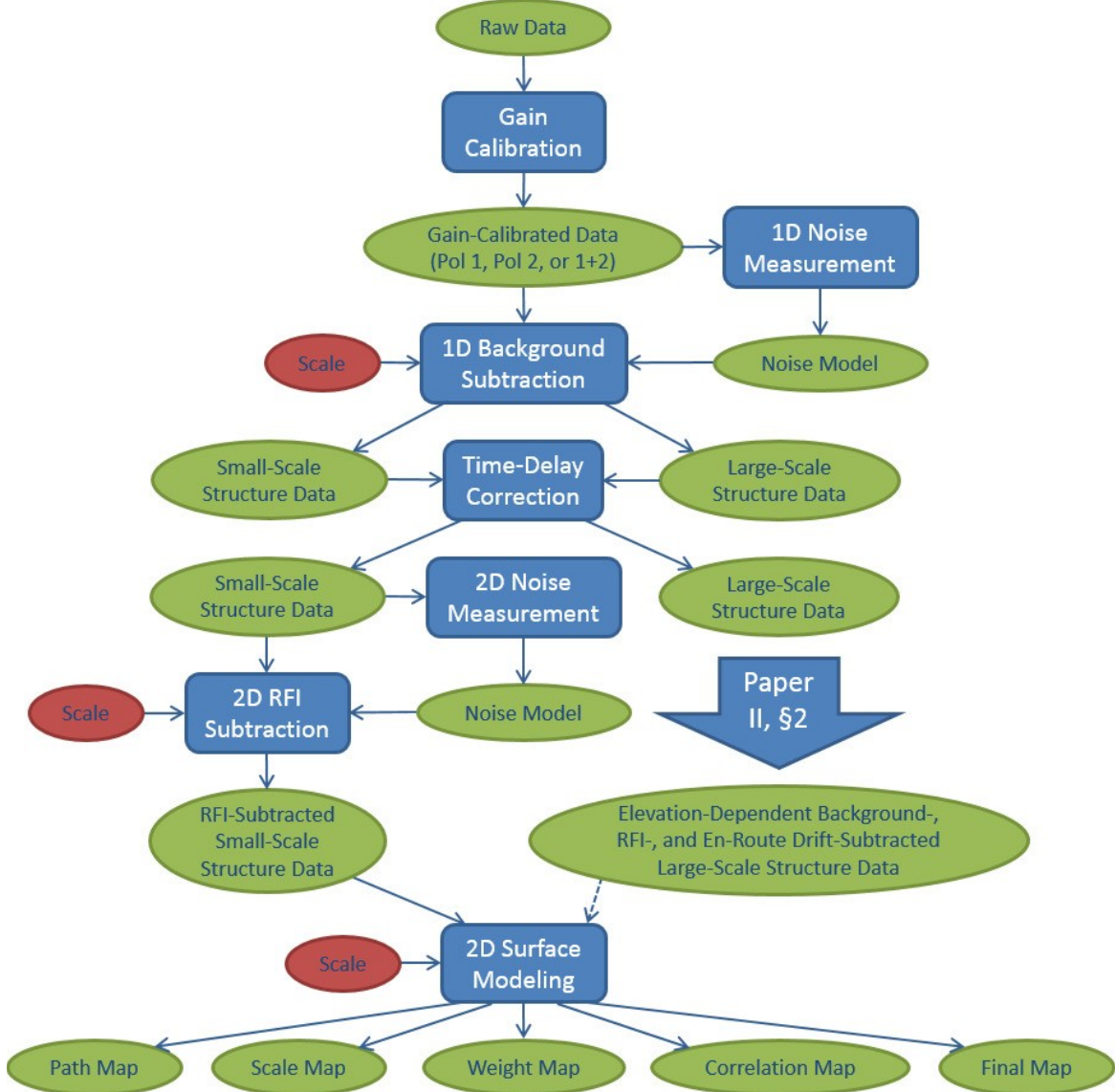


FIG. 7.— Flowchart of our algorithm for contaminant-cleaning and mapping small-scale structures. Blue are the component algorithms. Green are the inputs to and outputs of these component algorithms, consisting of data, corresponding noise models, and ultimately maps. Red are user-selected scales, for separating wanted and unwanted structures, and for modeling the final surface.

We approach all three types of signal contaminants in the same way. First, we model them locally, not globally: We use simple parameterizations, such as first- and second-order polynomials, but we do not expect, nor need, them to hold over large angular scales. When fitting these models, we use robust, and appropriate, outlier rejection (see §1.3) to separate contaminants from astronomical signal. Finally, we combine our locally-fitted models into a global model, checking the procedure against simulated data.

1.3. Robust Chauvenet Outlier Rejection

Our single-dish mapping algorithm is built upon a new outlier rejection technique, called “robust Chauvenet rejection”, which we developed for the Skynet Robotic Telescope Network’s image-processing library in general, and for this application in particular.

Sigma clipping (§1.2.3) is one of the simplest outlier rejection techniques, but also one of the crudest. It suf-

fes from a number of problems, foremost of which is how to set the threshold. For example, if working with ≈ 100 data points, 2-sigma variations are expected but 4-sigma variations are not. However, if working with $\approx 10^4$ data points, 3-sigma variations are expected but 5-sigma variations are not. Chauvenet rejection is simply sigma clipping plus a reasonable rule for setting the threshold:

$$NP(>|z|) = 0.5, \quad (1)$$

where N is the total number of data points and $P(>|z|)$ is the cumulative probability of being more than z standard deviations from the mean, assuming a Gaussian distribution (Chauvenet 1863).

However, sigma clipping, and by extension Chauvenet rejection, also suffers from the following problem: If the mean and the standard deviation are not known a priori, which is almost always the case, they must be measured from the data, and both of these quantities are sensitive to the very outliers that they are being used to reject.

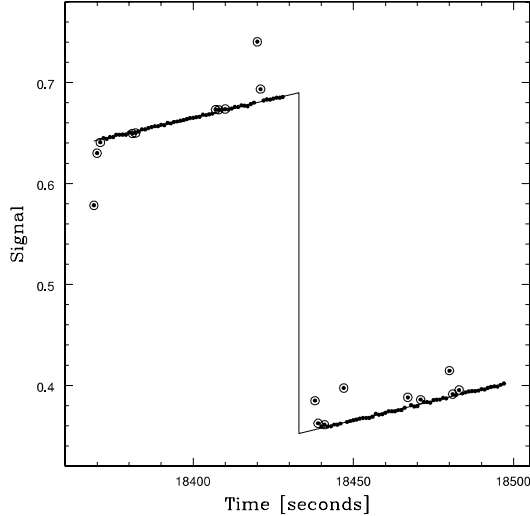


FIG. 8.— 40-foot gain calibration data, with the noise diode first on and then off, and best-fit model. Circled points have been robust-Chauvenet rejected, including data taken during the transitions from off to on and on to off, and RFI-contaminated data. The background level increased during the calibration, but our model accounts for this: Simply averaging each level, instead of modeling each level with a line, would have underestimated the result.

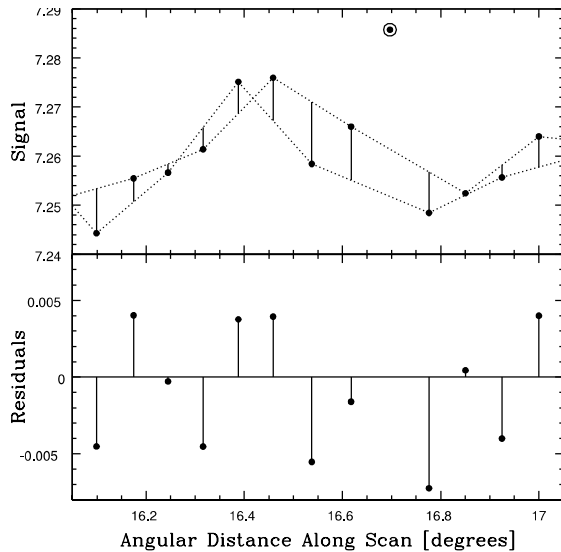


FIG. 9.— Point-to-point noise measurement technique. **Top:** Applied to gain-calibrated 20-meter data. The circled point has been robust-Chauvenet rejected. **Bottom:** Deviations. Mean and standard deviations are measured from the non-rejected points, for each scan.

This limits the applicability of Chauvenet rejection to very small sample-contamination fractions.

Consequently, we developed *robust* Chauvenet rejection, which makes use of the mean and the standard deviation, but also makes use of increasingly robust (but decreasingly precise) alternatives, namely the median and the (half-sample; e.g., Bickel & Fruhwirth 2005) mode, and the 68.3-percentile deviation, measured in three increasingly robust ways. These quantities approximate the mean and the standard deviation, respectively, and equal them in the case of a Gaussian distribution. But they are significantly less sensitive to outliers, and are

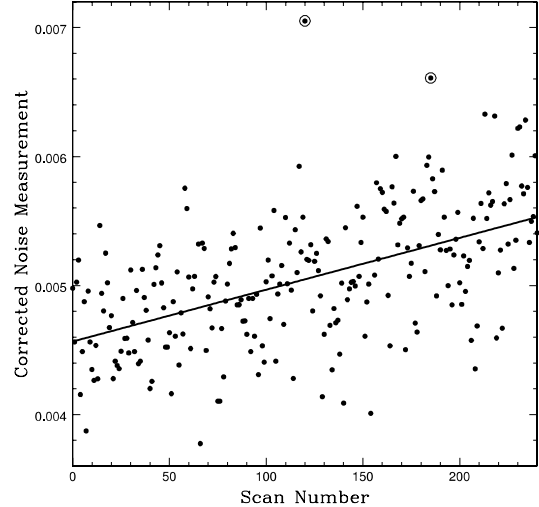


FIG. 10.— Corrected 1D noise measurements vs. scan number for a 20-meter observation, and best-fit model. Only two points met the robust Chauvenet rejection criterion, and then only barely, which is not unusual given that intra-scan outliers have already been rejected (Figure 9). The 1D noise level increased by $\approx 20\%$ over the course of this observation.

TABLE 1
MINIMUM RECOMMENDED 1D BACKGROUND-SUBTRACTION SCALE
FOR THE TELESCOPES AND RECEIVERS OF §2, IN THEORETICAL
BEAMWIDTHS

Telescope	Receiver	Scale
20-meter	L (HI + OH) ^a	7 ^c
20-meter	L (HI) ^b	6 ^c
20-meter	L (OH) ^b	6 ^c
20-meter	X	6 ^c
40-foot	L (HI)	3

^a Before August 1, 2014

^b After August 1, 2014

^c The 20-meter’s beam pattern has a low-level, broad component, in both L and X bands, and consequently, we recommend larger background-subtraction scales here. This component was significant in L band prior to 8/1/14, as can be seen in the third row of Figure 3, as well as in Figure 5. Post 8/1/14, it was significantly reduced, but not altogether eliminated. This component corresponds to approximately 2% – 3% and 4% – 5% of the integrated beam pattern in L and X band, respectively. If this is not a concern, these minimum recommended 1D background-subtraction scales can be lowered to 3 and 4 theoretical beamwidths, respectively.

particularly effective, meaning both robust and precise, if applied in proper combination.

The applicability of this technique is very broad, spanning not only astronomy and science, but all quantitative disciplines. We have submitted this technique as a companion paper (Maples et al. 2018), and make extensive use of it in this paper, with implementation details offered in the footnotes. As such, this paper additionally serves the purpose of “field testing” this new technique.

1.4. Overview

In §2, we provide a technical description of the refurbished 20-meter, including its L- and X-band receivers. We also provide a technical description of Green Bank Observatory’s 40-foot telescope, used primarily for education and public engagement, but on which we de-

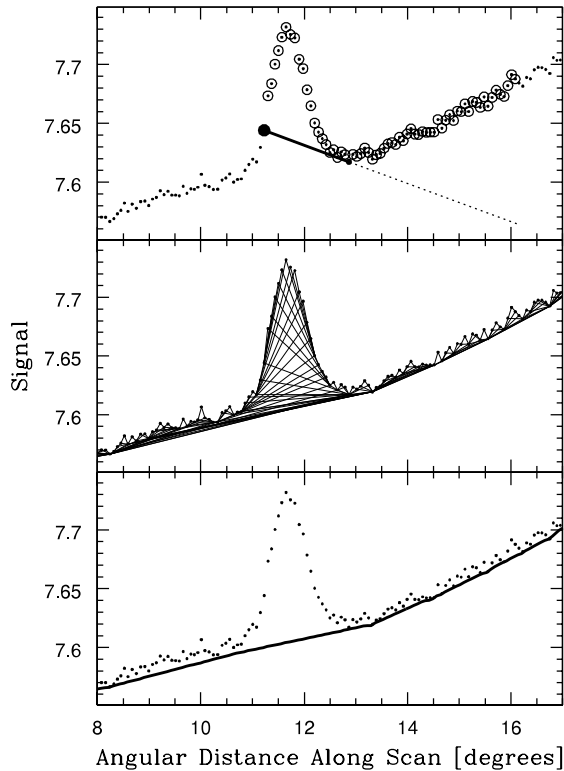


FIG. 11.— **Top:** A forward-directed local background model, anchored to an arbitrary point from Figure 5, near 3C 270. Circled points are within one background-subtraction scale length, but above the model. **Middle:** Forward- and backward-directed local background models, anchored to every point in the scan. **Bottom:** Global background model, constructed from the local background models.

veloped many components of our algorithm before the 20-meter was ready. In §3, we introduce our algorithm for contaminant-cleaning and mapping small-scale structures (e.g., sources) from continuum observations, and test it on both simulated and real data. In §4, we demonstrate that optical-style, aperture photometry can be carried out on these maps, and introduce an algorithm for calculating photometric error bars, given the different (and more complicated, correlated) noise characteristics of these maps. We summarize our results in §5.

In Paper II (Dutton et al. 2018): (1) We expand on our small-scale structure algorithm to additionally contaminant-clean and map large-scale structures; (2) We do the same for spectral (as opposed to just continuum) observations; and (3) We carry out an X-band survey of the Galactic plane, from $-5^\circ < l < 95^\circ$, to showcase, and further test, techniques developed in both papers.

2. TELESCOPES

2.1. Green Bank Observatory 20-Meter

The 20-meter diameter telescope (Figure 1) was constructed by Radiation Systems, Inc. at the Green Bank site, completed in 1994, and outfitted with a dual-frequency 2.3/8.4-GHz feed and receiver (Norrod et al. 1994). Funded by the U.S. Naval Observatory (USNO),

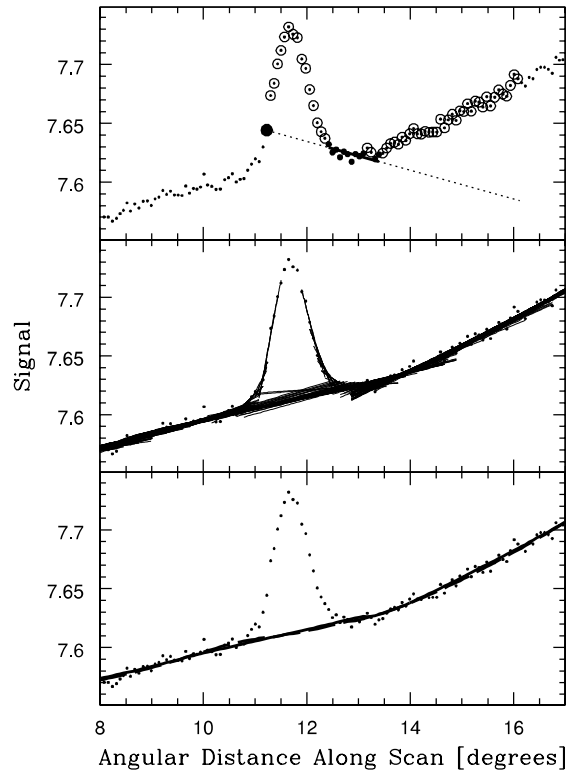


FIG. 12.— **Top:** A preliminary local background model, anchored to the same point as in Figure 11. Circled points have been iteratively rejected as too high, given the modeled noise level (§3.2); the larger points were not rejected. **Middle:** Final local background models, originally, but no longer, anchored to every point in the scan. **Bottom:** Global background model, constructed from the final local background models (solid curve), and from quadratic, instead of linear, local background models (dashed curve).

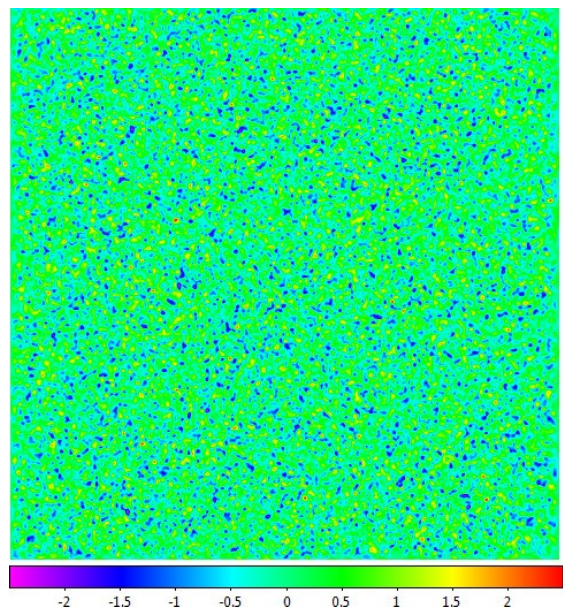


FIG. 13.— 20-meter 1/10-beamwidth horizontal raster replaced with Gaussian random noise, of mean zero and standard deviation one. Locally modeled surface (§1.2.1, see §3.7) has been applied for visualization only.

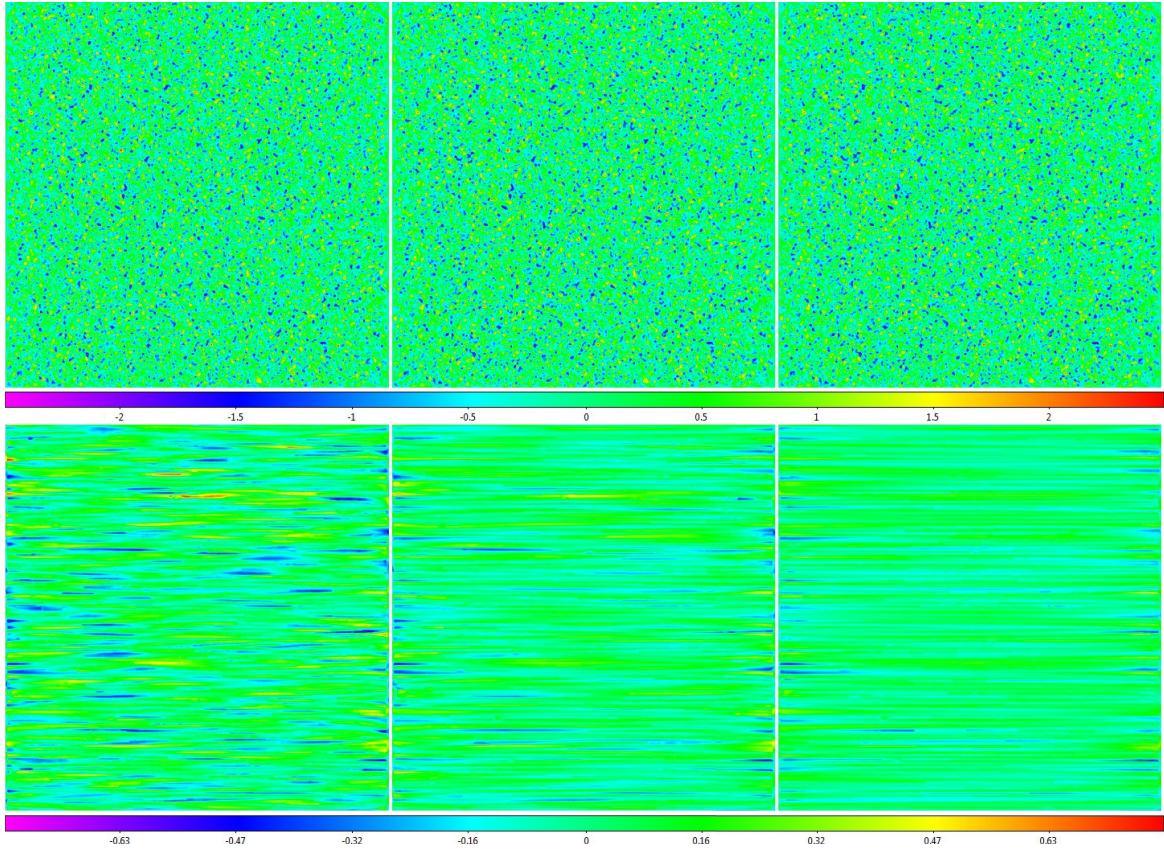


FIG. 14.— **Top Row:** Data from Figure 13 background-subtracted, with 6- (left), 12- (middle), and 24- (right) beamwidth scales (the map is 24 beamwidths across). **Bottom Row:** Data from the top row minus the data from Figure 13 (residuals). Background-subtracted data are not biased high nor low. To first order, the noise level of the background-subtracted data is $\approx 98.0\%$ (left), $\approx 98.8\%$ (middle), and $\approx 99.3\%$ (right) of that of the original data, and the RMS of the residuals is only $\approx 20.1\%$ (left), $\approx 15.4\%$ (middle), and $\approx 12.3\%$ (right) of the noise level of the original data (see Figure 15 for second-order effect). Locally modeled surfaces (§1.2.1, see §3.7) have been applied for visualization only.

it was part of a network of Very-Long Baseline Interferometry (VLBI) stations that monitored changes in Earth’s rotation axis and rate, as well as measured continental drift (Ma et al. 1998).

The antenna has an altitude/azimuth mount, and is able to observe down to 1 degree above the horizon. Its maximum slew speed is 2 degrees per second in each axis, which met the needs of the geodetic VLBI program, to quickly observe many radio sources, despite being widely distributed across the sky. Its pointing accuracy is ≈ 35 arcsec rms.

The antenna’s surface is a paraboloid with a focal ratio of 0.43 at prime focus. Its surface accuracy was originally 0.8 mm rms, but this has degraded over the years to ≈ 1.4 mm rms. Its aperture efficiency is $\approx 50\%$ at 8 GHz.

Due to budget cutbacks, USNO ended its Green Bank operations in 2000. From 2000 to 2010, the telescope was used only occasionally, primarily for testing experimental receivers.

In 2010, the University of North Carolina at Chapel Hill (UNC-CH) received a grant from the National Science Foundation to expand its Skynet Robotic Telescope Network, and this included restoring and automating the 20-meter.

Two receivers were prepared. The primary receiver is for 1.3 – 1.8 GHz (L band), and had previously been used on Green Bank’s 140-foot diameter telescope. We orig-

inally operated this receiver with three interchangeable feeds, each for a different part of the band. However, changing these feeds was a many-hour, manual operation. In 2014, we built and installed a new feed, for the entire 1.3 – 1.8 GHz band.

The secondary receiver is for 8 – 10 GHz (X band). It is a stripped-down version of the original, dual-frequency receiver, outfitted with a new 8 – 10 GHz feed. This receiver is used primarily when the L-band receiver is off the telescope for maintenance, which has only been for two extended (many-month) periods since 2010.

A new backend was also prepared. The spectrometer uses a Field-Programmable Gate Array (FPGA) design that was adapted from the Green Bank Ultimate Pulsar Processing Instrument (GUPPI). In its low resolution mode, it produces 1024-channel spectra over a 500-MHz band. In 2013, a high-resolution mode was added; in this mode, it produces 1024-, 2048-, 4096-, 8192-, or 16286-channel spectra in each of two 15.625-MHz bands (see §3 of Paper II).

New software was also written for low-level control of (1) the antenna’s pointing, and (2) data acquisition from the spectrometer, as well as for high-level control of the entire system through Skynet’s web-based interface (Footnote 7). For both its optical telescopes and its radio telescope, Skynet features a dynamically updated queue, with options for assigning different users, groups

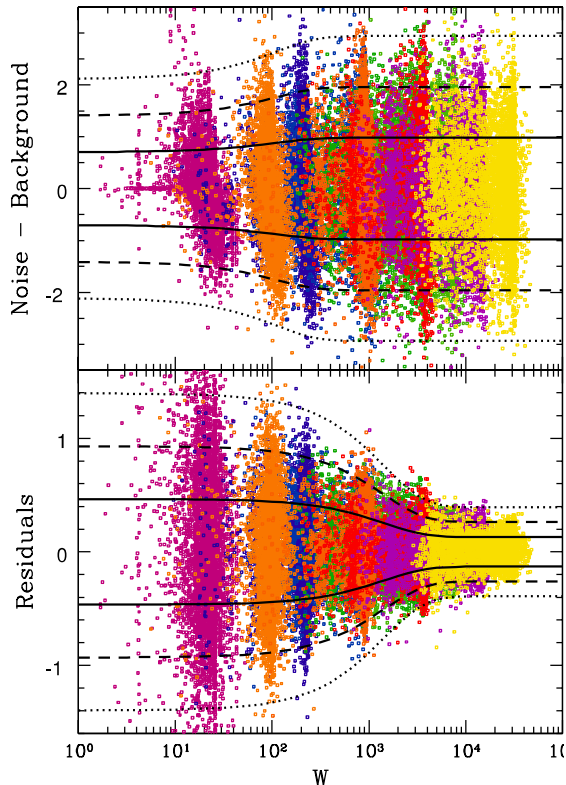


FIG. 15.— Gaussian random noise background-subtracted (top) and residuals (bottom) vs. W , the sum of the weights of the non-rejected local background models that determine the global background model at each point, for 1/10- and 1/5-beamwidth rasters, and for 1-, 3-, 6-, 12-, and 24-beamwidth background-subtraction scales (background-subtraction scale times sampling density of the data sets increases from left to right). The RMS of the data varies with W , but not with background-subtraction scale or sampling density independently. Curves are 1-, 2-, and 3- σ model noise envelopes that have been fitted to all of these data simultaneously (Equations 6 and 7).

of users, and collaborations of groups different levels of priority access, including target-of-opportunity access.

Beyond the initial grant, continued operation of the 20-meter has been made possible by a combination of education grants and research grants (primarily to search for fast radio bursts), as well as by operations funds and personnel of the Green Bank Observatory. Since the 20-meter went online on Skynet in 2012, over 23,000 observations have been carried out, mostly by students, for both education and research objectives.

2.2. Green Bank Observatory 40-Foot

The 40-foot diameter telescope (see Figure 6) was obtained in 1961 to determine if several bright radio sources varied on daily timescales.

Purchased from Antenna Systems, Inc., this relatively inexpensive (\$40K) meridian-transit telescope was prefabricated in Hingham, Massachusetts, and assembled at the Green Bank site in two days.

The reflecting surface is aluminum mesh, similar to that of Green Bank’s former 300-foot diameter telescope, but with smaller openings. The reflector backup structure is also aluminum, with a supporting pedestal of galvanized steel. The telescope’s maximum slew speed is 20 degrees per minute along the local meridian, and it can point far enough south to observe the Galactic center,

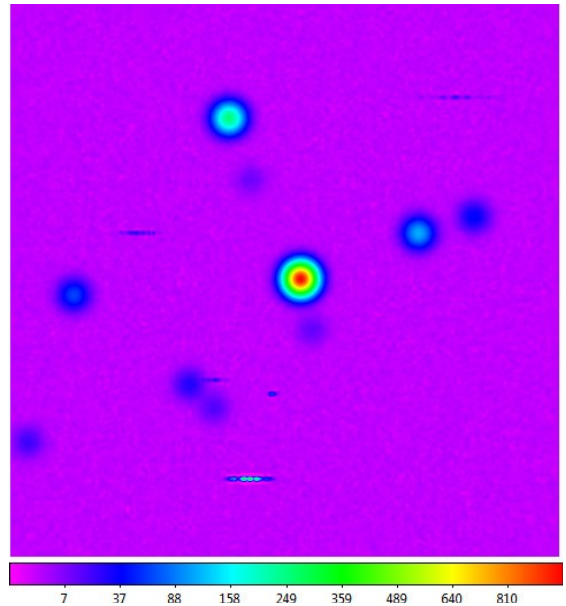


FIG. 16.— Simulated data from Figure 13 to which we have added point sources and short-duration RFI. For the point sources, we use a Gaussian beam pattern. For the short-duration RFI, we use the absolute value of a sum of rapidly varying sine functions multiplied by a short-duration Gaussian envelope function. Locally modeled surface (§1.2.1, see §3.7) has been applied for visualization only. Square-root scaling is used to emphasize fainter structures.

and as far north as a few degrees beyond the celestial pole.

The original receivers and feeds were built onsite, and operated at 750 and 1400 MHz. Backend electronics are housed in an adjacent, underground bunker, to prevent radio-frequency noise from the electronics from interfering with celestial signals.

The original electronics automatically repointed the telescope to the declination of each source sufficiently before it transited the meridian each day. Data were digitized and written out to punched paper tape. Put into operation in 1962, the 40-foot is thought to have been the world’s first fully automated telescope (Bowyer & Meredith 1963; Lockman, Ghigo, & Balser 2007).

This project ran for several years, but was abandoned because none of the sources varied on daily timescales. Afterward, they were instead observed less frequently, but with higher precision, using Green Bank’s 300- and 140-foot diameter telescopes.

The 40-foot fell into disuse for about twenty years. But in 1987, it was refurbished as an educational resource, for students from 5th grade through graduate school to learn basic observational radio astronomy. The feed was replaced with one that had been used on Green Bank’s 85-foot diameter Tatel telescope for Project OZMA, sensitive to 18 – 21 cm. A new receiver was built at the National Radio Astronomy Observatory’s Central Development Laboratory in Charlottesville, Virginia. Spectral-line capability was added using a synthesizer (which has since been upgraded, see below) and narrowband filters that were inherited from Green Bank’s 300-foot diameter telescope after its collapse in 1988.

Over the past thirty years, several generations of students, numbering in the high thousands, have used the 40-foot as part of their science education. For peda-